



## CONTACT

+49 1796905749

[anniebhalla16@gmail.com](mailto:anniebhalla16@gmail.com)

[LinkedIn: /anniebhalla](https://www.linkedin.com/in/anniebhalla)

[github.com/Anniebhalla16](https://github.com/Anniebhalla16)

## SKILLS

### Robotics & Perception

ROS2, SLAM, 3D Gaussian Splatting, stereo depth estimation, sensor fusion, computer vision, camera calibration, hyperspectral imaging, OpenCV, Open3D

### ML Frameworks

PyTorch 2.1, CUDA 12.1, torchvision, torchmetrics

### Programming

Python, C++, TypeScript

### Systems & Infrastructure

Docker, GitHub Actions  
CI/CD, RabbitMQ, REST APIs, FASTapi, PostgreSQL, MongoDB

## LANGUAGES

German (B2) [\[B1 Certificate\]](#),  
English (native), Hindi  
(native)

# Annie Bhalla

AI SPACE ROBOTICS · ANALOG ASTRONAUT · SOFTWARE ENGINEER

## / 01 · WORK EXPERIENCE

07/2026

– present

### Flight Control Team Member

[Austrian Space Forum — AMADEE-27 analog Mars mission](#)

- Authoring Data Management Workflow Definitions and Guidelines, formalising data and communication interfaces across Flight Crew, Ground Operations and Mission Support Center.
- Defined interface specifications (data type, medium, timing, cadence) per team pairing, following ICD conventions per ECSS practice.
- Contacts Officer (in-training), Flight Control Team — coordinating real-time information flow between MSC console positions during simulation.

01/10/2025

– present

### Software Engineer

[Sereact, Stuttgart, Germany](#)

- Improved KPI reliability from 50% to ~95% by developing telemetry and real-time analytics software for 100+ robot stations, enabling performance tracking and data-driven decision-making across locations.
- Designed scalable event-driven telemetry analysis pipelines on RabbitMQ, WebSockets and a microservices architecture, ingesting and evaluating real-time robot telemetry across timezones.

01/02/2026

– 08/02/2026

### Analog Astronaut (Extra Vehicular Activity Lead - EVAL)

[Aaka Space Studio – Lunar Surface Simulation, India · Certificate](#)

- Led 3 EVAs over an 8-day lunar analog mission, coordinating supply pick-ups and scientific data collection at points of interest on the Dholavira terrain, following mission timelines, risk assessments and procedural checklists.

19/10/2025

– 28/10/2025

### Mission Control Officer — Remote Science Support Task Manager

[Austrian Space Forum — World's Biggest Analog Mission, Vienna · Certificate](#)

- Coordinated remote scientific operations and managed data integrity checks for 17 globally distributed habitat analog sites over 8 days.

01/11/2024

– 01/10/2025

### Research Student — Master Thesis - DLR-RMC

[Deutsches Zentrum für Luft- und Raumfahrt \(DLR\), Oberpfaffenhofen](#)

- Designed and validated a visual navigation pipeline integrating hyperspectral and RGB-D depth estimation with 3DGS SLAM for planetary surface navigation in unstructured extraterrestrial terrain.
- Built automated evaluation pipeline benchmarking trajectory error across algorithmic parameter sweeps — achieved 42% ATE reduction (6.95 cm → 5.05 cm) over RGB-only baseline.
- Conducted field data acquisition on [Moon-Mars Outdoor Test Site](#) at DLR Oberpfaffenhofen; managed ROS2-based multi-sensor data synchronisation and validation across sensor modalities.
- Thesis accepted for presentation at the [International Astronautical Congress \(IAC\) 2026](#).

## / 02 · EDUCATION

04/2023

– 10/2025

### MSc Computer Science – Autonomous Systems (1.8)

[\[Transcript\]](#)

[University of Stuttgart](#)

Thesis: HyperLoop — Hyperspectral-Assisted Loop Closure in 3DGS SLAM for Planetary Exploration

## / 03 · ACHIEVEMENTS & CERTIFICATIONS

04/2026

Interactive Presentation, International Astronautical Congress IAC 2026  
[\(Notification Letter\)](#)

06/2026

Analog Mission Basic Training [Certification](#), Austrian Space Forum

07/2025 – 10/2025

Recipient of DAAD STIBET Scholarship [Certificate](#)

06/2025 – 07/2025

Asclepius Analog Astronaut Selection Program Phase D Candidate (last round) [Certificate](#)